

Mukund Mangla

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[LinkedIn](#) • [Github](#) • [Leetcode](#) • [Personal Portfolio](#)



rishihood
university



Newton School
of Technology

PROFESSIONAL SUMMARY

Curious engineer driven by the challenge of turning bold ideas into real, impactful products — blending code, design, and AI to craft experiences that feel futuristic, practical, and genuinely different from the ordinary. Constantly building, learning, and pushing boundaries beyond conventional development

EDUCATION

Bachelor of Technology (Artificial intelligence)
Newton School of Technology, Rishihood University

2024 - 2028
Grade: 7.0/10.0

Intermediate (Class XII)
Tagore Public School

2022 - 2023
Grade: 76.0%

Matriculation (Class X)
Tagore Public School

2020 - 2021
Grade: 80.0%

PROJECTS

GriefCompanion ([Github](#)) ([Demo](#))

March 2026

- **Tech Stack:** React.js, Node.js, Express.js, Python, Flask, MongoDB Atlas, Groq API (LLaMA 3.1), ChromaDB, TF-IDF Retrieval, DistilBERT, HuggingFace Transformers, Clerk Authentication, Vercel, Render
- **Description:** An AI-powered emotional writing assistant that helps people compose the hardest messages of their lives — condolences, apologies, difficult news, and reconnections — through a premium, distraction-free interface backed by a full RAG pipeline and real-time LLM generation.
- **Features:** Secure user authentication using Clerk, AI-powered letter generation using Groq's LLaMA 3.1 with real-time token streaming via Server-Sent Events, Retrieval-Augmented Generation (RAG) pipeline using TF-IDF semantic search over a curated corpus of human-written letters, fine-tuned DistilBERT emotion classifier for auto-detecting message intent, multi-turn feedback and refinement loop allowing users to critique and regenerate drafts up to 3 times, tone A/B comparison generating two parallel variants using Promise.all, MongoDB Atlas persistence for draft history with soft-delete and edit tracking, admin analytics dashboard built on MongoDB aggregation pipelines visualised with Recharts, session auto-save restoring form state across browser refreshes, graceful degradation across all failure modes ensuring the app never crashes

Ember ([Github](#)) ([Demo](#))

March 2026

- **Tech Stack:** React.js, Node.js, Express.js, MongoDB (Mongoose), Socket.io (WebSockets), JWT (JSON Web Tokens), Context API & Custom Hooks, Framer Motion, Lucide Icons, CSS Variables (Bespoke Cream & Amber theme), Vercel, Render
- **Description:** A premium, full-stack real-time restaurant management and order tracking platform designed to streamline cafe and restaurant operations through a modern, responsive interface, live visual table tracking, and a dedicated kitchen display pipeline.
- **Features:** Secure user authentication using JWT and bcryptjs password hashing with role-based routing, real-time bi-directional order processing and status synchronization across all devices via Socket.io, interactive visual table layout management tracking live table states (Available, Occupied, Reserved) with drag-and-drop table transfer functionality, a dedicated Kitchen Display System (KDS) featuring smart prep timers and order priority visual flags, advanced checkout billing with dynamic bill splitting (by amount or items) and invoice receipt generation, a persistence-backed shift handover message board allowing team-wide coordination between shifts, an executive analytics dashboard leveraging MongoDB aggregation pipelines to calculate revenue, peak hours, and popular dishes visualized with high-fidelity custom CSS bar charts and heatmaps, session auto-save and automated socket reconnection protecting state against sudden disconnections, and strict security middleware including Express Rate Limit, Helmet headers, and Mongo Sanitize protection against NoSQL injection.

habitflow ([Github](#)) ([Demo](#))

March 2026

- **Tech Stack:** React.js, Node.js, Express, MongoDB, Google Gemini AI, Zustand, JWT Authentication
- **Description:** An AI-powered habit tracking operating system designed to gamify personal growth through structured routines, rich analytics, and intelligent insights.

- **Features:** Secure custom JWT authentication, comprehensive habit tracking with category-based organization and visual analytics, gamified user progression including XP, leveling, and streak systems, a beautifully minimal and responsive UI featuring smooth micro-animations, and an integrated AI assistant powered by Google Gemini for personalized habit coaching and routine generation.

Balancio ([Github](#)) ([Demo](#))

December 2025

- **Tech Stack:** React 19, Node.js, Express.js, MongoDB, Mongoose ORM, JWT, Socket.io, Vanilla CSS
- **Description:** Collaborative fintech platform for managing shared expenses and simplified debt settlements, designed to streamline financial interactions within social groups through real-time synchronization and intuitive visualization.
- **Features:** Real-time expense synchronization via Socket.io ensuring instant balance updates across all connected clients, smart debt simplification algorithm reducing transaction complexity by minimizing required payments, interactive Bento Grid dashboard with dynamic spending visualization, simplified split logic supporting equal and custom distribution, secure dual-layer JWT authentication with persistent sessions, Zustand for optimized state management reducing re-renders and strictly typed with TypeScript, fully responsive mobile-first custom CSS design (320px–1920px) achieving 95 lighthouse performance score, comprehensive activity feed for transparent audit logging of all financial actions, 'Nudge' notification system for settlement reminders, and robust schema validation using Zod for ensuring data integrity across financial transactions.

Ops-Pilot ([Github](#)) ([Demo](#))

November 2025

- **Tech Stack:** React 19, Node.js, Express.js, MongoDB, Mongoose ORM, JWT, Tailwind CSS
- **Description:** Enterprise internal operations platform for managing approval workflows and structured request handling with SLA-based tracking, immutable audit logging, and role-based access control.
- **Features:** Multi-stage approval workflows with strict assignee enforcement, dynamic request templates with schema validation, SLA-driven overdue detection via cron-based monitoring, immutable audit logs for traceability, dual-layer JWT authentication with role-based authorization, TanStack Query caching reducing redundant API calls by 65%, Zustand-powered state persistence with session recovery, rate-limited request creation, mobile-responsive Tailwind UI (320px–1920px), operational analytics dashboard derived from audit logs, and a scalable roadmap including escalation automation and WebSocket notifications.

CERTIFICATIONS

Cybersecurity Analyst Job Simulation Tata (Link)	May 2026
Cybersecurity Analyst Job Simulation	
Data Analytics Job Simulation Deloitte (Link)	May 2026
Data Analytics Job Simulation	

SKILLS

AI / ML: Transformers, LangChain, OpenAI API, GenAI, LangGraph, Hugging Face, PyTorch, MCP, Gemini API, Vector Embeddings, Deep Learning (DL), RAG, Machine Learning, n8n Introduction, Scikit-learn, Natural Language Processing, TensorFlow, LLM, Generative AI

APIs & Testing: Postman, ESLint, API, API testing

Cloud & DevOps: Git and Github, GitHub Actions

Computer Languages: SQL, TypeScript, HTML, CSS, JavaScript, Python

Courses: Data Structure

Databases & Data Tools: PostgreSQL, Excel, Tableau, MySQL, MongoDB, NoSQL, Supabase

Domain & Business Skills: Canva, UI/UX, Product Management, Figma, Analytics, SEO/SEM marketing, Entrepreneurship

Frameworks & Libraries: Next.js, Tailwind CSS, Framer Motion, Three.js, FastAPI, Next JS, Prisma ORM, NumPy, React, Node.js, SciPy, Express JS, Pandas, Chart.js, Tailwind, Microsoft Office

Soft Skills: Problem-Solving, Communication Skills, Interpersonal communication, Public speaking, Written communication, Leadership, Team Building, Time management, Decision-making, Creativity, Teamwork, Critical Thinking, Openness, People skills